

David Rollo

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M.S. Applied Mathematics candidate with professional experience in NLP, backend optimization, and algorithmic design. Proven track record of improving system performance and deploying production-grade Python/Go solutions.

Education

University of Utah | *M.S. Applied Mathematics*

Aug 2024 - Present

Certificate: Deep Learning, Artificial Intelligence and Robotics

- Relevant Coursework: High-dimensional data analysis, optimization, reinforcement learning, and graph theory.

Utah State University | *B.S. Computational Mathematics*

Aug 2019 - May 2023

Minors: Anticipatory Intelligence, Statistics

- Capstone: Analyzed Dept. of Energy data to model threat resilience in the US electric grid, applying methods from Anticipatory Intelligence including modelling dynamic system interactions and risk forecasting.

Industry Experience

Signals | *Software Engineer*

March 2025 - Present

- Processed a dataset of 2.5 million audio recordings and transcriptions using topic modeling to isolate high-value data clusters for model training.
- Engineered backend optimizations in PostgreSQL, achieving a 60x reduction in query latency and a 30% increase in vector-based retrieval accuracy.
- Automated quality control systems to flag lead qualification in AI chat conversations.

OutlierAI | *Code Review and SME*

Nov 2023 - Jan 2025

- Evaluated and refined RLHF training data for LLMs, auditing code quality across 6 metrics to ensure numerical accuracy and reasoning consistency.
- Corrected complex SQL queries joining between 15+ tables generated from client models.

James Madison University + NSF | *Applied Researcher*

May 2022 - Oct

2022

- Automated data pipelines to generate networks with variable connectivities, compare results with previous methods, and unit-test our parallelizable, unsupervised algorithm.
- Co-authored and presented a research paper on directed graph community detection at MIT's URTC conference; open-sourced the Python preprocessing algorithm on GitHub.

Relevant Skills

Languages — Python, Java, TypeScript, Go, R

Data Science — PyTorch, Pandas, NumPy, Scikit-Learn, Regression, Neural networks

Web & Tools — Git, Bash, APIs, React, HTMX, PostgreSQL